A CONTINUOUS RIGHT ADJOINT OF A COCONTINUOUS FUNCTOR DEFINES THE SAME MORPHISM OF TOPOI

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The morphism of topoi associated to a cocontinuous functor and the description of its functors. Same chapter, Lemma lemma-have-functor-otherway, recorded as the context of this card. Supplies the morphism of topoi and the identifications used in the proof.
- The definition of a morphism of sites. Same chapter, Definition definition-morphism-sites. Supplies the exactness condition that has to be checked for the right adjoint.

Source reference. Original lemma 00XO: sources/stacks-sites/pinned-original-source.tex, source lines 5063–5063. Navigation pointer invoked at proof line 5063; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 00X1: sources/stacks-sites/pinned-original-source.tex, source lines 5109–5109. Navigation pointer invoked at proof line 5109; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be sites. Let $u : \mathcal{C} \to \mathcal{D}$, and $v : \mathcal{D} \to \mathcal{C}$ be functors. Assume that u is cocontinuous and that v is a right adjoint to u. Let $g : Sh(\mathcal{C}) \to Sh(\mathcal{D})$ be the morphism of topoi associated to u, see Lemma 00XO. Then*

1. *for a sheaf $\mathcal{F}$ on $\mathcal{C}$ the sheaf $g_*\mathcal{F}$ is equal to the presheaf $v^p\mathcal{F}$, in other words, $(g_*\mathcal{F})(V) = \mathcal{F}(v(V))$, and*
2. *for a sheaf $\mathcal{G}$ on $\mathcal{D}$ we have $g^{-1}\mathcal{G} = (v_p\mathcal{G})^\#$.*

Lemma 0.2. *Notation and assumptions as in Lemma 0.1. If in addition v is continuous then v defines a morphism of sites $f : \mathcal{C} \to \mathcal{D}$ whose associated morphism of topoi is equal to g.*

Proof. We will use the results of Lemma 0.1 without further mention. To prove that v defines a morphism of sites f as in the statement of the lemma, we have to show that v_s is an exact functor (see Definition 00X1). Since $v_s\mathcal{G} = (v_p\mathcal{G})^\# = g^{-1}\mathcal{G}$ this follows from the fact that g is a morphism of topoi. Then we see that $f^{-1} = v_s = g^{-1}$ and we find that $f = g$ as morphisms of topoi. □